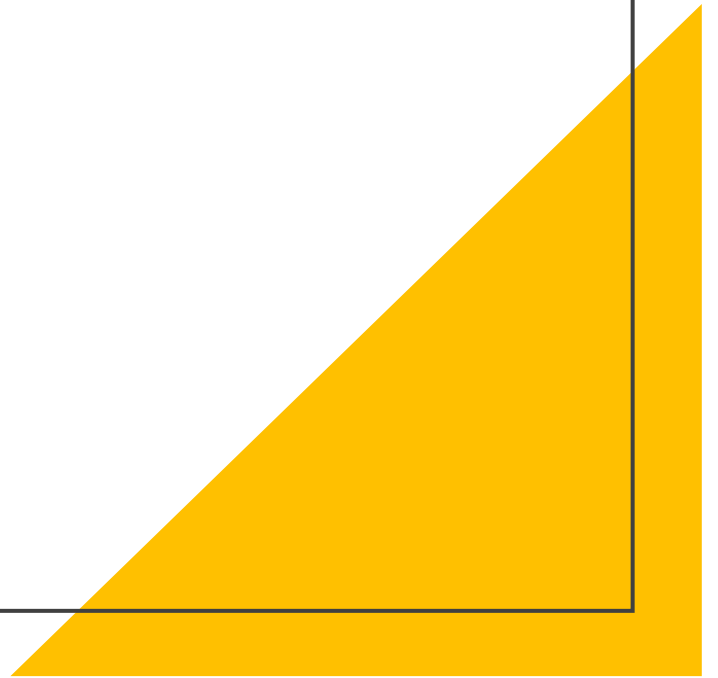


Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 9 – Security Warning Design II



Last time

- Three takeaways from early research on security warnings
 - Passive warnings are not obsolete: active and passive warnings play different roles
 - Warnings need to be usable
 - Warning utility is contextual
- Discussion grounded in early browser warnings
 - What should warnings in LLMs look like
 - Maybe we need to communicate different information than in phishing contexts
 - Maybe it depends on the importance of the task
 - Difference between the intention of passive LLM warnings and browser warnings
 - Ethics questions around phishing & warning experiments
 - Do lab studies hold up to all phishing threat models?

Today

- Discuss specific design problems in warnings
 - Salah will be presenting and leading discussion on one of the papers
- Examples of early security warnings results applied to recent problems
 - Social media mis/disinformation warnings
 - Results from a 2021 paper
 - Synthetic content & social media warnings for kids
 - Recent tech + policy discussions and open research problems

Browser security warnings are a success story

Browser warnings broadly deployed across all major web browser

Browser warnings protect users

- Safe Browsing assess **more than 10 billion URLs** and files daily & shows **more than 3 million user warnings** ([Google 2025](#))

Browser warnings continue to be worked on

- LLM-generated and explainable warnings
- Warnings in new settings (e.g., should warnings be different as new forms of authentication get adopted?)
 - Warnings are contextual

Mis/disinformation warnings



Facebook 2016 fact-check warning



Facebook COVID-19 misinformation warning



Subscribe

Have you ever wondered why the Wikimedia Foundation wants so much money?

It certainly isn't needed to operate Wikipedia. You can literally fit a copy of the entire text on your phone!

So, what's the money for? Inquiring minds want to know ...

Readers added context they thought people might want to know

The Wikimedia Foundation is a charitable non-profit, providing free access to Wikipedia. While a text and English-only copy of Wikipedia is about 51GB, including all media and supported languages totals to 428TB. In 2022 Wikimedia had \$154M in revenue with \$145M in expenses. en.wikipedia.org/wiki/Wikipedia... wikimediafoundation.org/about/annualre...

X community notes

Drawing on lessons from early warning research

Problem: There was little evidence that warnings help users identify or avoid disinformation

- 10-15 years earlier: usable security and privacy researchers grappled with the **same problem** in a different context (phishing & malware)

Lessons researchers in later decades took away:

1. Active warnings (interstitial) heeded more than passive (contextual)
2. Effects of habituation on users' behavior
3. Approach to studying warnings
 1. Measuring behavioral metrics (e.g., click through rate)
 2. Task-focused lab studies

Warnings for Online Disinformation

Empirically evaluate interstitial and contextual disinformation warnings

1. Will users notice and understand the warnings?
2. Will users change their behavior after seeing the warnings?
3. What messaging strategies are most effective at changing user behavior?

Part 1: Laboratory experiment

- Laboratory experiment (n = 40)
- Think aloud + interview
 - 4 search tasks using Google Search and Chrome
 - 2 control rounds & 2 treatment rounds with either contextual or interstitial warnings
 - Participants told to use one of two specific websites
 - Participants told to start with first website then could switch to the second
 - First source carried a warning
- Measures:
 - Clickthrough rate
 - Alternative visit rate (AVR)
 - Observation notes

Find the political party of Taiwan's Premier on TheNewsLens or FocusTaiwan.

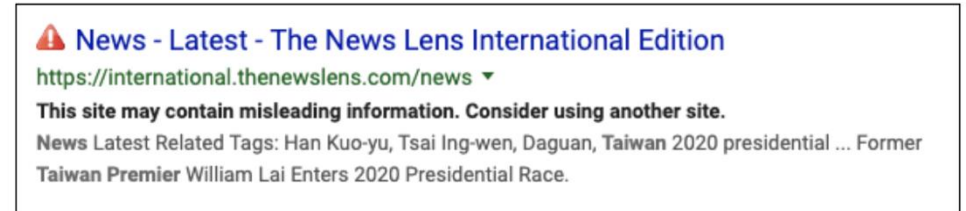


Figure 1: Search results pages displayed contextual warnings.

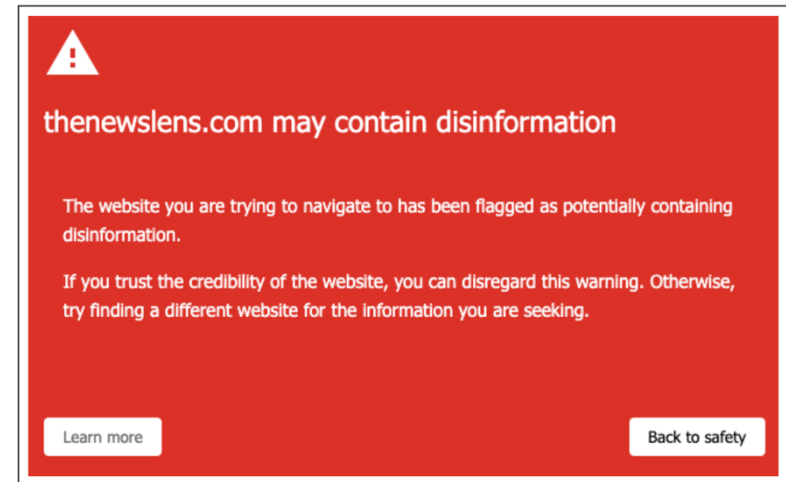


Figure 2: Participants encountered interstitial warnings after clicking search results.

Results

- CTR for contextual warnings was higher than for interstitial warnings
 - 33/40 v. 18/40
- 13/20 participants in the contextual treatment group were not aware they had been shown disinformation warnings
- All 20 participants noticed the interstitial warnings
 - 12 understood these were about misinformation
 - 7 thought they were about a risk of harm, a virus, or another security threat
- Factors influencing behavior changes:
 - Informativeness
 - Fear of harm
 - Friction

Part 2: Crowdworker study

- 250 participants
- 8 interstitial disinformation warnings
 - 4 designed for informativeness
 - 4 designed to evoke fear of harm
- Participants completed search tasks online
 - Used an application that simulated a web browser with pre-populated results
 - Target search results contained answer
 - Nontarget search results did not
- Measures:
 - Pre-task partisanship survey
 - CTR & AVR
 - Decoy survey about why they chose results
 - Perception survey after treatment rounds
- Control rounds had no warning
- Treatment rounds presented warning on first target results participants clicked



False or Misleading Content Warning

This website presents itself as news, but it contains information that experts have identified to be false or misleading

This website spreads disinformation: lies, half-truths, and non-rational arguments intended to manipulate public opinion.

It can be difficult to tell the difference between real news and disinformation, but it poses a serious threat to national security, election integrity, and democracy.

Go back

Dismiss and continue



Security Alert

This website is dangerous.

Consider finding alternative sources of information.

Go back

Dismiss and continue

Results

- Confirmed results from part 1: interstitial warnings have a strong effect on user behavior
- Interstitial warnings can inform users and make them think about trustworthiness of websites
 - Did not find evidence that informative warnings have greater effect on behavior than non-informative
- Same results for warnings that evoke fear
 - Warning design can convey risk of harm, but inconclusive on whether better conveying this affects user behavior
- No evidence that partisanship moderates warning perceptions or behaviors

Warnings on AI and social media

- Synthetic media
- Warnings for children
- LLM permissions

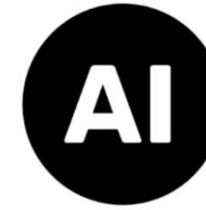
Synthetic media

- Some governments have required labels or warnings on AI-generated content
 - EU AI Act: labels on a range of AI-generated or modified content
 - CA: disclosure on video/audio ads that use AI-generated performers
- Social media AI labels
 - RQs:
 - Rate of mislabeling?
 - Impact on content creators?

Basic icon

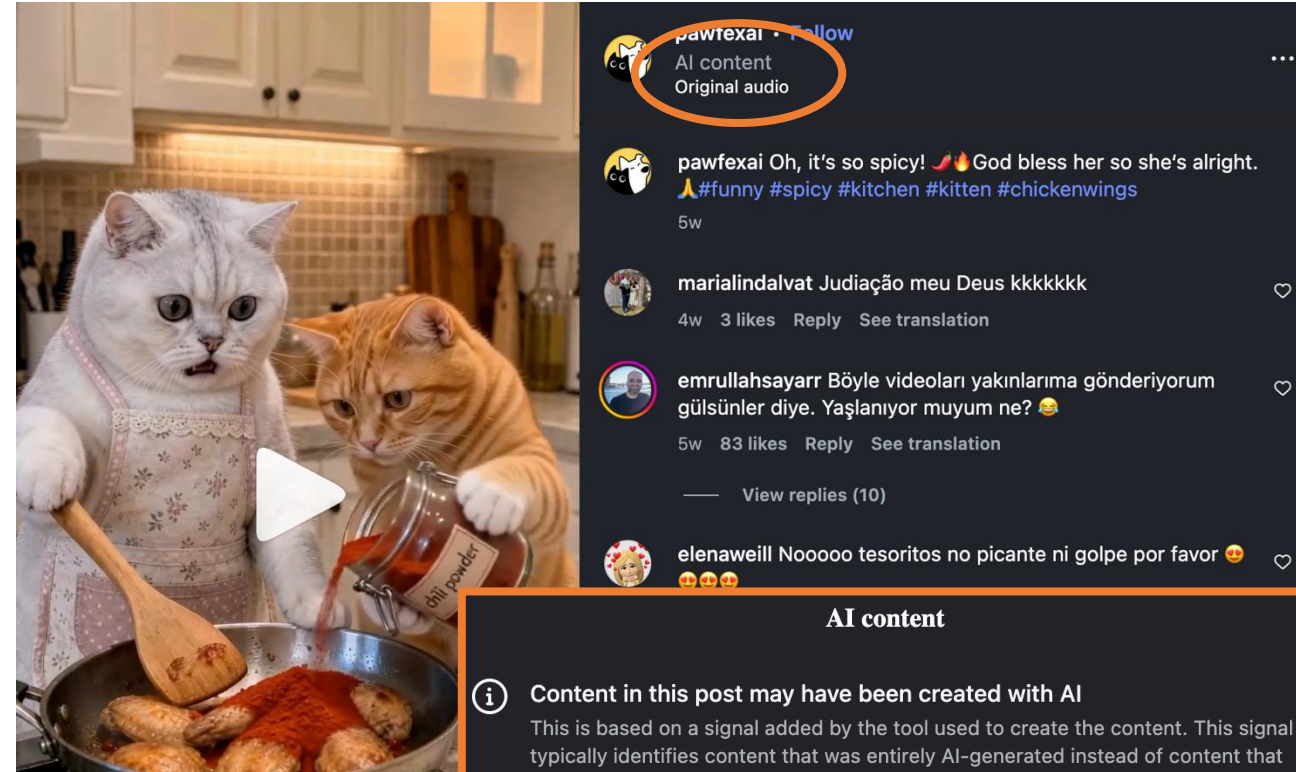
Fully AI-Generated

Partially AI-Modified



AI GENERATED

AI MODIFIED



AI content

- i Content in this post may have been created with AI**
This is based on a signal added by the tool used to create the content. This signal typically identifies content that was entirely AI-generated instead of content that was only modified with AI.
- 📄 Helping you identify AI content**
We add a label when a signal indicates that content is entirely AI-generated. Not all AI content contains the data we need to identify it. We share information about AI use in the post's menu. [Learn more](#)

Warnings for children on social media



- Concerns around children's use of social media are prevalent
 - Minnesota and California recently passed bills requiring social media warning labels

“The Surgeon General has warned that while social media may have benefits for some young users, social media is associated with significant mental health harms and has not been proven safe for young users.”

- Open problems:
 - Can social media warnings actually change young people's habits?
 - How should the labels be designed?
 - E.g., should they be more informative? Should they evoke fear?
 - What kind of interaction should they require?
 - What is the right time to show users a warning label?

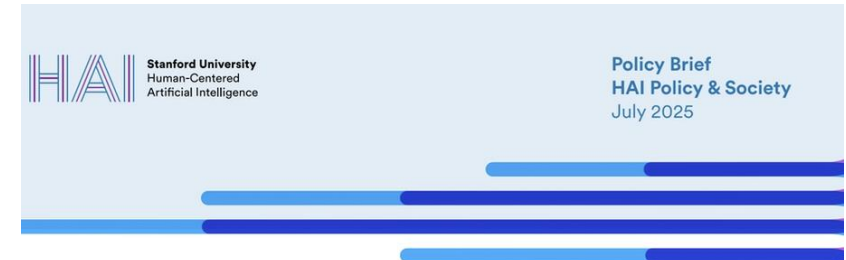
LLM chatbot and agent permissions

Current state:

- LLM chatbots present passive warnings to users
 - In the EU, these are mandated by law
- AI agents don't broadly contain warnings for security risks
 - Do contain permissions

Lots of open problems:

- On chatbots:
 - How do AI labels empower users to make more informed decisions about content consumption?
- On agents:
 - How should we warn users about various security risks?
 - Overreach
 - Cascading failures in multi-agent systems
 - Prompt injection attacks
 - Sensitive data exposure



HAI | Stanford University
Human-Centered
Artificial Intelligence

Policy Brief
HAI Policy & Society
July 2025

Labeling AI-Generated Content May Not Change Its Persuasiveness

Isabel O. Gallegos, Chen Shani, Weiyan Shi, Federico Bianchi, Izzy Gainsburg, Dan Jurafsky, and Robb Willer

GENERATIVE AI TOOLS CAN NOW PRODUCE PERSUASIVE CONTENT AT PREVIOUSLY UNPRECEDENTED SCALE AND SPEED. There are many ways in which these tools can be used for positive impact in the world. But the emergence of persuasive, AI-generated content also makes possible many negative uses — such as influence operations, misinformation campaigns, and other kinds of deception — particularly in political contexts. These risks are compounded by a key issue: People struggle to distinguish AI-generated content from content written by humans, which helps influence campaigns and misinformation thrive.

These risks have led policymakers to call for authorship labels on AI-made content. In the European Union, for instance, the AI Act requires that entities deploying AI-generated or -manipulated content label it as such. In the United States, the AI Labeling Act and the AI Disclosure Act, both introduced in Congress in 2023 but not passed, would have implemented similar rules. Policymakers' calls for labels on AI-made content raise a key question: Will a label change how much the content influences people's political and public policy views?

Key Takeaways

In response to the rapidly improving ability of AI tools to create persuasive content, policymakers are increasingly calling for labels on AI-generated content—but little research has measured whether adding a label impacts the persuasiveness of the underlying messages.

We surveyed how more than 1,500 people perceive AI-generated policy messages when told the content had been created by an expert AI model, a human policy expert, or told nothing about its authorship.

Adding the label changed people's perceptions of whether the author was AI or human but did not significantly change the persuasiveness of the content itself, regardless of the policy domain (e.g., allowing colleges to pay student athletes) or participant demographics (e.g., political party).

Policy proposals requiring AI content labels may enhance transparency, but their inability to affect persuasiveness highlights the need for complementary safeguards (e.g., media literacy education) and ongoing research into how AI disclosure policies shape the information ecosystem.

Next time

- Privacy on Monday
 - Final day of usable security and privacy
 - Akshat presenting and leading discussion
- I will be adding grades to Canvas in the next week or two, including participation grade so far
- Continue submitting assignments on Gradescope